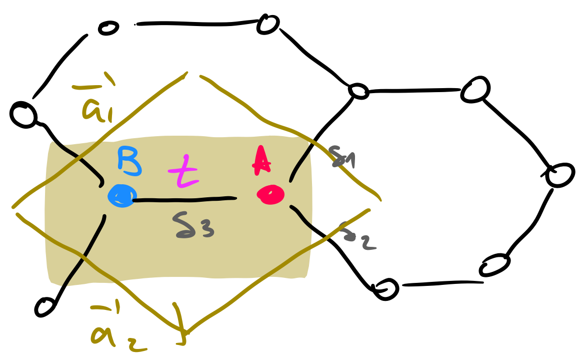
$\boxed{P1}$ 2D graphene:



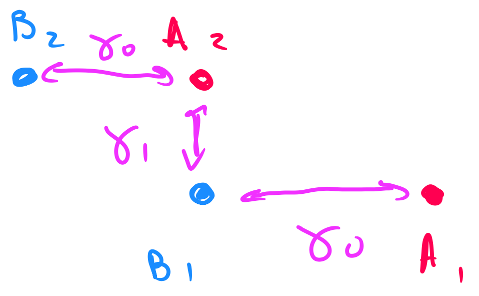
$$H = \sum_k' \psi^\dagger(k)\, h(k)\, \psi(k)$$

$$h(k) = \begin{pmatrix} 0 & f(k) \\ f^*(k) & 0 \end{pmatrix}; \quad f(k) = -t \sum_s' e^{i\vec{k}\vec{s}} \text{ – hopping ampl}$$

A-B stacking

$$s_1 = \frac{a}{2}(1, \sqrt{3}); \; s_2 = \frac{a}{2}(1, -\sqrt{3}), \; s_3 = \frac{a}{2}(-2, 0)$$

$$h(k) = \begin{pmatrix} 0 & f_{A_1B_1} & 0 & 0 \\ f^*_{A_1B_1} & 0 & f_{A_2B_1} & 0 \\ 0 & f^*_{A_2B_1} & 0 & f_{A_2B_2} \\ 0 & 0 & f^*_{A_2B_2} & 0 \end{pmatrix}$$



$\tilde{a} = 3.55$ Å,
$a = 1.42$ Å
$\gamma_2 = 3.03$ eV
$\gamma_2 = 0.39$ eV

$$h(q) = \begin{bmatrix} 0 & \frac{3at}{2}(q_x - iq_y) & 0 & 0 \\ \frac{3at}{2}(q_x + iq_y) & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & \frac{3at}{2}(q_x - iq_y) \\ 0 & 0 & \frac{3at}{2}(q_x + iq_y) & 0 \end{bmatrix}$$

b)
$$\det(H_{bi}(q) - EI) = \begin{array}{c|cccc} & A_1 & B_1 & A_2 & B_2 \\ \hline A_1 & -\varepsilon & f & 0 & 0 \\ B_1 & f^* & -\varepsilon & \gamma & 0 \\ & 0 & \gamma & -\varepsilon & f \\ & 0 & 0 & f^* & -\varepsilon \end{array} =$$

$$= -\varepsilon \begin{vmatrix} -\varepsilon & \gamma & 0 \\ \gamma & -\varepsilon & f \\ 0 & f^* & -\varepsilon \end{vmatrix} - f \begin{vmatrix} f^* & \gamma & 0 \\ 0 & -\varepsilon & f \\ 0 & f^* & -\varepsilon \end{vmatrix} = -\varepsilon(-\varepsilon^3 + f^2\varepsilon + \gamma^2\varepsilon) - f(f^*\varepsilon^2 - f^2 f^*) =$$

$$= \varepsilon^4 - (f^2 + \gamma^2)\varepsilon^2 - f^2\varepsilon^2 + f^4 = \varepsilon^4 - (2f^2 + \gamma^2)\varepsilon^2 + f^4 = (\varepsilon^2 + \gamma\varepsilon - f^2)(\varepsilon^2 - \gamma\varepsilon - f^2) = 0$$

$$\varepsilon_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 + 4f^2}}{2}; \quad \varepsilon_{3,4} = \frac{\gamma^2 \pm \sqrt{\gamma^2 - 4f^2}}{2}$$

Using notation from the problem:

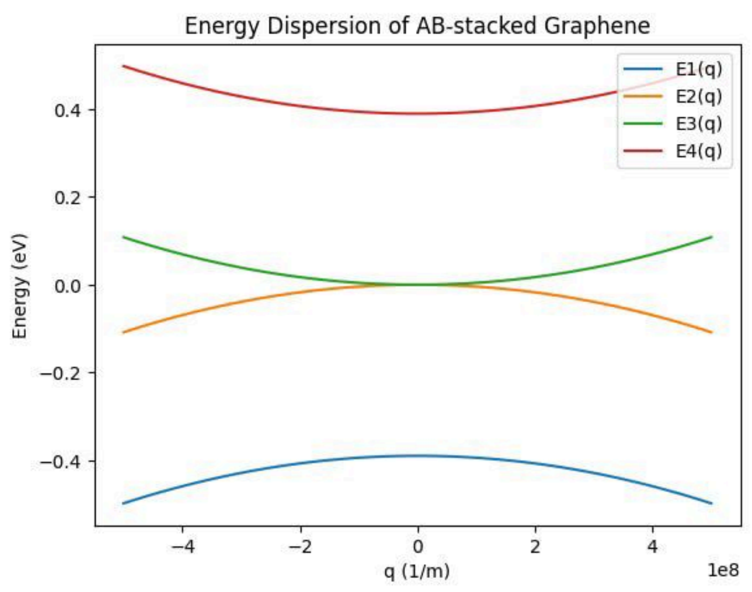
$$\varepsilon_{1,2,3,4} = \frac{\pm\gamma_1 \pm \sqrt{\gamma_1^2 + 4(\gamma_0 S(k))^2}}{2}$$

c) $\gamma_0 S(k) = \hbar v q$

$$\varepsilon_1 = \frac{-\gamma_1 - \gamma_1\sqrt{1 + 4\frac{(\hbar v q)^2}{\gamma_1}}}{2} \simeq$$

$$\simeq -\frac{\gamma_1}{2}\left(1 + 1 + \frac{2(\hbar v q)^2}{\gamma_1}\right) = -\gamma_1 - (\hbar v q)^2$$

$$\varepsilon_2 \simeq -(\hbar v q)^2$$



d) External field breaks the symmetry: A_1 and B_1 are no longer equivalent, so the bandgap opens

Matrix multiplication is analogous to part(b), & did it in python

$$h = \begin{vmatrix} -\frac{V}{2} & f - V/2 & 0 & 0 \\ f^* - \frac{V}{2} & -V/2 & \gamma & 0 \\ 0 & \gamma & +V/2 & f \\ 0 & 0 & f^* & +V/2 \end{vmatrix}; \quad f \simeq \hbar v_F q$$